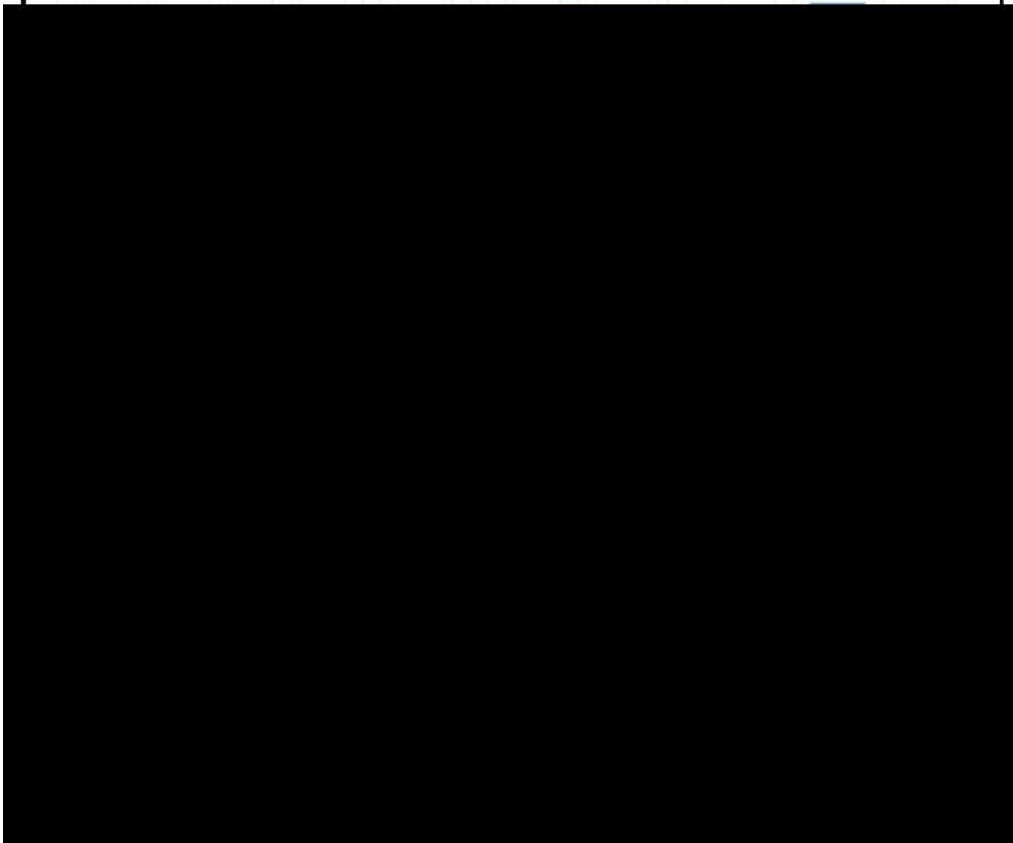
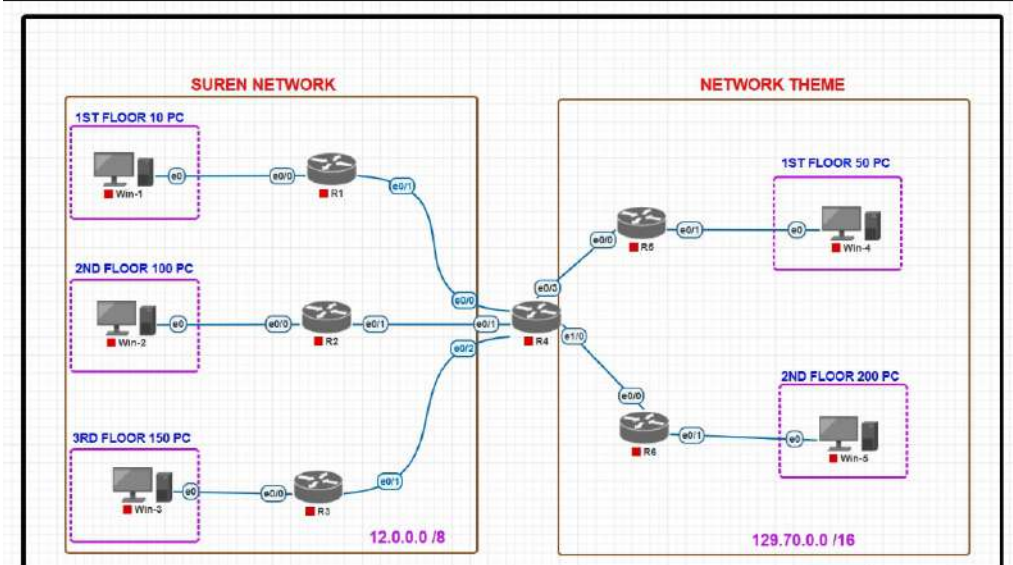
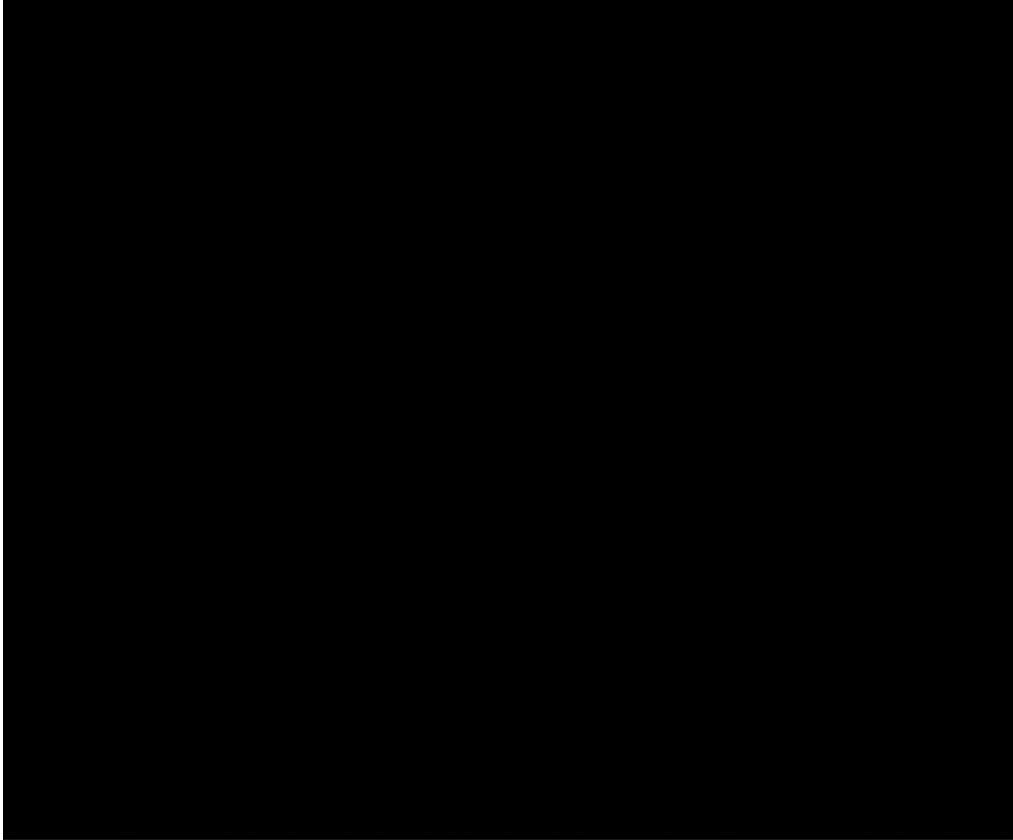




```
R1
R1#show ip int br
Interface              IP-Address      OK? Method Status    Prot
ocol
Ethernet0/0            12.0.1.129      YES manual up        up
Ethernet0/1            12.0.1.145      YES manual up        up
Ethernet0/2            unassigned      YES TFTP  down      down
Ethernet0/3            unassigned      YES TFTP  down      down

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.146
S       12.0.1.0/25 [1/0] via 12.0.1.146
C       12.0.1.128/28 is directly connected, Ethernet0/0
L       12.0.1.129/32 is directly connected, Ethernet0/0
C       12.0.1.144/30 is directly connected, Ethernet0/1
L       12.0.1.145/32 is directly connected, Ethernet0/1
S       12.0.1.148/30 [1/0] via 12.0.1.146
S       12.0.1.152/30 [1/0] via 12.0.1.146
    129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       129.70.0.0/24 [1/0] via 12.0.1.146
S       129.70.1.0/26 [1/0] via 12.0.1.146
S       129.70.1.64/30 [1/0] via 12.0.1.146
S       129.70.1.68/30 [1/0] via 12.0.1.146
R1#
R1#
R1#
...▲■
```

```

R2#show ip int br
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              12.0.1.1        YES manual up            up
Ethernet0/1              12.0.1.149      YES manual up            up
Ethernet0/2              unassigned      YES TFTP  down          down
Ethernet0/3              unassigned      YES TFTP  down          down
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.150
C       12.0.1.0/25 is directly connected, Ethernet0/0
L       12.0.1.1/32 is directly connected, Ethernet0/0
S       12.0.1.128/28 [1/0] via 12.0.1.150
S       12.0.1.144/30 [1/0] via 12.0.1.150
C       12.0.1.148/30 is directly connected, Ethernet0/1
L       12.0.1.149/32 is directly connected, Ethernet0/1
S       12.0.1.152/30 [1/0] via 12.0.1.150
    129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       129.70.0.0/24 [1/0] via 12.0.1.150
S       129.70.1.0/26 [1/0] via 12.0.1.150
S       129.70.1.64/30 [1/0] via 12.0.1.150
S       129.70.1.68/30 [1/0] via 12.0.1.150
R2#
R2#
R2#
R2#
R2#

```

R3

```
R3(config)#exit
R3#
*Sep 22 08:05:37.675: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip int br
Interface                               IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0                             12.0.0.1        YES manual up          up
Ethernet0/1                             12.0.1.153      YES manual up          up
Ethernet0/2                             unassigned      YES TFTP  down        down
Ethernet0/3                             unassigned      YES TFTP  down        down

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
C       12.0.0.0/24 is directly connected, Ethernet0/0
L       12.0.0.1/32 is directly connected, Ethernet0/0
S       12.0.1.0/25 [1/0] via 12.0.1.154
S       12.0.1.128/28 [1/0] via 12.0.1.154
S       12.0.1.144/30 [1/0] via 12.0.1.154
S       12.0.1.148/30 [1/0] via 12.0.1.154
C       12.0.1.152/30 is directly connected, Ethernet0/1
L       12.0.1.153/32 is directly connected, Ethernet0/1
    129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       129.70.0.0/24 [1/0] via 12.0.1.154
S       129.70.1.0/26 [1/0] via 12.0.1.154
S       129.70.1.64/30 [1/0] via 12.0.1.154
S       129.70.1.68/30 [1/0] via 12.0.1.154
R3#
R3#~
```

```
R4
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    12.0.1.146      YES manual up          up
Ethernet0/1    12.0.1.150      YES manual up          up
Ethernet0/2    12.0.1.154      YES manual up          up
Ethernet0/3    129.70.1.66     YES manual up          up
Ethernet1/0    129.70.1.70     YES manual up          up
Ethernet1/1    unassigned      YES unset  administratively down down
Ethernet1/2    unassigned      YES unset  administratively down down
Ethernet1/3    unassigned      YES unset  administratively down down

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 9 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.153
S       12.0.1.0/25 [1/0] via 12.0.1.149
S       12.0.1.128/28 [1/0] via 12.0.1.145
C       12.0.1.144/30 is directly connected, Ethernet0/0
L       12.0.1.146/32 is directly connected, Ethernet0/0
C       12.0.1.148/30 is directly connected, Ethernet0/1
L       12.0.1.150/32 is directly connected, Ethernet0/1
C       12.0.1.152/30 is directly connected, Ethernet0/2
L       12.0.1.154/32 is directly connected, Ethernet0/2
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S       129.70.0.0/24 [1/0] via 129.70.1.69
S       129.70.1.0/26 [1/0] via 129.70.1.65
C       129.70.1.64/30 is directly connected, Ethernet0/3
L       129.70.1.66/32 is directly connected, Ethernet0/3
C       129.70.1.68/30 is directly connected, Ethernet1/0
L       129.70.1.70/32 is directly connected, Ethernet1/0
Router#
```

R5

```
R5(config)#exit
R5#
*Sep 22 08:06:29.379: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip int br
Interface                IP-Address      OK? Method Status  Protocol
Ethernet0/0              129.70.1.65     YES manual up      up
Ethernet0/1              129.70.1.1      YES manual up      up
Ethernet0/2              unassigned      YES TFTP  down    down
Ethernet0/3              unassigned      YES TFTP  down    down
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       12.0.0.0/24 [1/0] via 129.70.1.66
S       12.0.1.0/25 [1/0] via 129.70.1.66
S       12.0.1.128/28 [1/0] via 129.70.1.66
S       12.0.1.144/30 [1/0] via 129.70.1.66
S       12.0.1.148/30 [1/0] via 129.70.1.66
S       12.0.1.152/30 [1/0] via 129.70.1.66
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S       129.70.0.0/24 [1/0] via 129.70.1.66
C       129.70.1.0/26 is directly connected, Ethernet0/1
L       129.70.1.1/32 is directly connected, Ethernet0/1
C       129.70.1.64/30 is directly connected, Ethernet0/0
L       129.70.1.65/32 is directly connected, Ethernet0/0
S       129.70.1.68/30 [1/0] via 129.70.1.66
R5#
R5#
```

```
R6
R6(config)#exit
R6#
*Sep 22 08:07:18.811: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip int br
Interface                               IP-Address      OK? Method Status      Prot
-----
Ethernet0/0                             129.70.1.69      YES manual up          up
Ethernet0/1                             129.70.0.1       YES manual up          up
Ethernet0/2                             unassigned       YES TFTP  down        down
Ethernet0/3                             unassigned       YES TFTP  down        down

R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       12.0.0.0/24 [1/0] via 129.70.1.70
S       12.0.1.0/25 [1/0] via 129.70.1.70
S       12.0.1.128/28 [1/0] via 129.70.1.70
S       12.0.1.144/30 [1/0] via 129.70.1.70
S       12.0.1.148/30 [1/0] via 129.70.1.70
S       12.0.1.152/30 [1/0] via 129.70.1.70
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
C       129.70.0.0/24 is directly connected, Ethernet0/1
L       129.70.0.1/32 is directly connected, Ethernet0/1
S       129.70.1.0/26 [1/0] via 129.70.1.70
S       129.70.1.64/30 [1/0] via 129.70.1.70
C       129.70.1.68/30 is directly connected, Ethernet0/0
L       129.70.1.69/32 is directly connected, Ethernet0/0
R6#
R6#
R6#
```



```
PuTTY (inactive)
R1(config)#
R1(config)#
R1(config)#
R1(config)#ip dhcp pool 1ST Floor
R1(dhcp-config)#network 12.0.1.128 255.255.255.240
R1(dhcp-config)#default-router 12.0.1.129
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#
R1(config)#ip dhcp excluded-address 12.0.1.129 12.0.1.134
R1(config)#exit
R1#
R1#
R1#
R1#
*Sep 22 06:03:16.087: %SYS-5-CONFIG_I: Configured from console by console
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#ip dhcp pool 2nd Floor
R1(dhcp-config)#network 12.0.1.0 255.255.255.128
R1(dhcp-config)#default-router 12.0.1.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#
R1(config)#ip dhcp excluded-address 12.0.1.1 12.0.1.10
R1(config)#exit
R1#
R1#
*Sep 22 06:17:38.494: %SYS-5-CONFIG_I: Configured from console by console
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#ip dhcp pool 3rd floor
R1(dhcp-config)#network 12.0.0.0 255.255.255.0
R1(dhcp-config)#default-router 12.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#
R1(config)#ip dhcp excluded-address 12.0.0.1 12.0.0.10
R1(config)#exit
```


PUTTY (inactive)

```
R1#
R1#
*Sep 22 06:24:15.814: %SYS-5-CONFIG_I: Configured from console by console
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 129.70.1.0 255.255.255.192
R1(dhcp-config)#default-router 129.70.1.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 129.70.1.1 129.70.1.10
R1(config)#exit
R1#
R1#
*Sep 22 06:31:36.621: %SYS-5-CONFIG_I: Configured from console by console
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#ip dhcp pool SALES
R1(dhcp-config)#network 129.70.0.0 255.255.255.0
R1(dhcp-config)#default-router 129.70.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#
R1(config)#ip dhcp excluded-address 129.70.0.1 129.70.0.10
R1(config)#exit
R1#
R1#
```

 PuTTY (inactive)

```

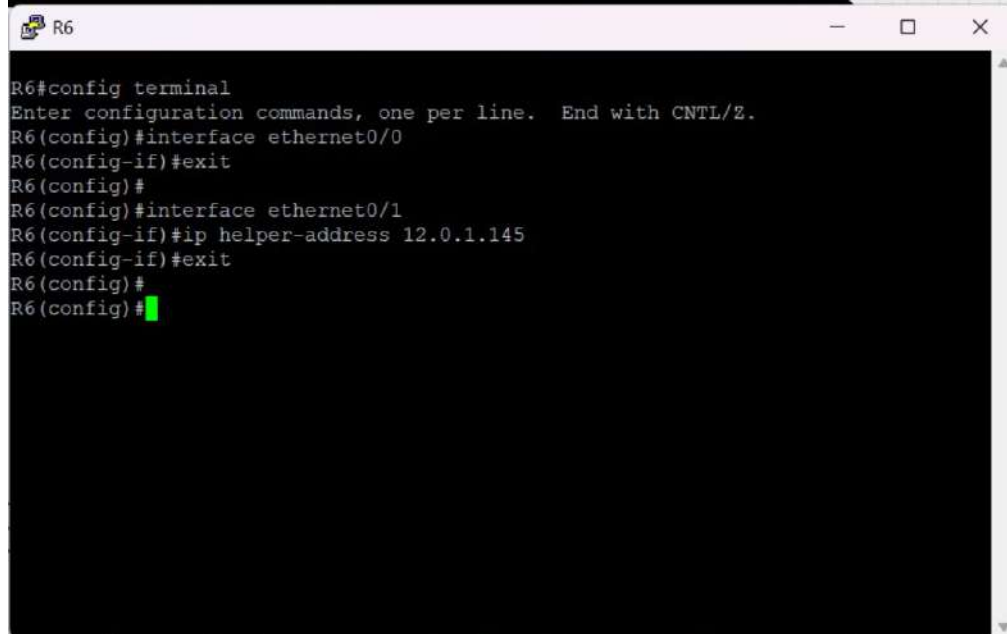
Pool 1ST Floor :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 14
Leased addresses                  : 1
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.1.136         12.0.1.129 - 12.0.1.142      1

Pool 2nd Floor :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 126
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.1.1           12.0.1.1 - 12.0.1.126      0

Pool 3rd floor :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.0.1           12.0.0.1 - 12.0.0.254      0

Pool HR :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 62
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
129.70.1.1         129.70.1.1 - 129.70.1.62      0

Pool SALES :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Pending event                     : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
    
```



```
R6#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#interface ethernet0/0
R6(config-if)#exit
R6(config)#
R6(config)#interface ethernet0/1
R6(config-if)#ip helper-address 12.0.1.145
R6(config-if)#exit
R6(config)#
R6(config)#
```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 12.0.1.135
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 12.0.1.129

C:\Documents and Settings\user>ping 12.0.1.150

Pinging 12.0.1.150 with 32 bytes of data:

Reply from 12.0.1.150: bytes=32 time=2ms TTL=254
Reply from 12.0.1.150: bytes=32 time=1ms TTL=254
Reply from 12.0.1.150: bytes=32 time=1ms TTL=254
Reply from 12.0.1.150: bytes=32 time=1ms TTL=254

Ping statistics for 12.0.1.150:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 129.70.1.69

Pinging 129.70.1.69 with 32 bytes of data:

Reply from 129.70.1.69: bytes=32 time=1ms TTL=253
Reply from 129.70.1.69: bytes=32 time=1ms TTL=253
Reply from 129.70.1.69: bytes=32 time=1ms TTL=253
Reply from 129.70.1.69: bytes=32 time=1ms TTL=253

Ping statistics for 129.70.1.69:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

